

Quantum Machine Learning with Neural Networks

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Outline

Quantum Machine Learning Landscape

- Quantum Data and Classical Data: $|\psi\rangle$ vs vectors $\mathbf{x} \in \mathbb{R}^n$
- Categories:
 - Quantum-enhanced classical ML
 - Quantum-native ML algorithms
 - Hybrid quantum-classical ML
- Resource-aware quantum learning models

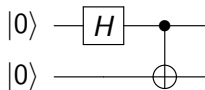
Quantum States and Gates

- Qubit: $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, $|\alpha|^2 + |\beta|^2 = 1$
- Common Gates:

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \quad X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

- Entanglement via CNOT and multi-qubit systems

Quantum Circuit Example



- Bell state: $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$

Classical Neural Networks Review

- Universal approximation theorem
- Layer-wise structure: $y = \sigma(Wx + b)$
- Deep networks and expressivity
- Limitations in high-dimensional feature space

What is a QNN?

- Quantum analog of classical NNs using PQCs
- Data encoding + entangling + variational layer
- Output via observable measurements

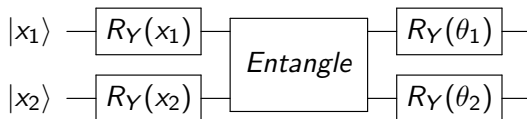
Generic QNN Architecture

Figure: QNN: Feature map + variational layers + measurement

Variational Quantum Circuits

- PQCs with parameters θ
- Optimize: $C(\theta) = \langle \psi(\theta) | \hat{H} | \psi(\theta) \rangle$

Circuit Example: Ansatz Layer



- Gradient via Parameter Shift Rule:

$$\frac{\partial \langle O \rangle}{\partial \theta} = \frac{\langle O \rangle_{\theta+\pi/2} - \langle O \rangle_{\theta-\pi/2}}{2}$$

Applications of QML + NN

- Quantum-enhanced classification
- QGANs, Quantum autoencoders
- Anomaly detection, Quantum RL

Challenges in QNNs

- Hardware limitations, barren plateaus
- Hybrid training instability

Future Directions

- Efficient hybrid architectures
- Adaptive QNNs and better ansätze

Thank You!

Questions?

- Inspired by classical CNNs for hierarchical feature extraction.
- Layers include quantum convolution and pooling operations.
- Reduce qubit counts while preserving quantum information.

QCNN Example Circuit

- Quantum convolution applies parameterized gates to pairs of qubits.
- Pooling reduces the system size, e.g., via measurement or entanglement filtering.

Figure: Illustrative QCNN circuit with convolution and pooling layers.

Quantum GAN Framework

- Combines quantum generator and classical or quantum discriminator.
- Generator learns to produce quantum states matching target distribution.
- Applications: quantum state preparation, data augmentation.

QGAN Architecture

Figure: Example QGAN setup with a quantum generator and classical discriminator.

Benchmarks in QML Research

- Performance metrics: classification accuracy, fidelity, loss convergence.
- Datasets: quantum-enhanced MNIST, quantum chemistry datasets.
- Notable results:
 - QNN classifiers outperform classical on small quantum datasets.
 - QCNNs demonstrate improved generalization with fewer qubits.
 - QGANs achieve state fidelity $\geq 95\%$ on target distributions.

Example Benchmark Table

Model	Dataset	Accuracy (%)	Qubits Used
QNN	Iris (Quantum Enc.)	94.3	4
QCNN	Quantum MNIST	89.7	6
QGAN	Quantum States	95.5 (Fidelity)	5

Table: Sample benchmark results for QNN, QCNN, and QGAN models.